```
Microsoft Windows [Version 10.0.26100.7309]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>CD\

C:\>CD XAMPP

C:\xampp>cd mysql

C:\xampp\mysql>cd bin


C:\xampp\mysql\bin>mysql -u root -p -h 127.0.0.1
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 9
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

1) Create any database name

MariaDB [(none)]> CREATE DATABASE lab5_dbms;

Query OK, 1 row affected (0.002 sec)

```
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| customer_db        |
| eemc_db            |
| information_schema |
| lab5_dbms          |
| mysql              |
| nitish_db          |
| performance_schema |
| phpmyadmin         |
| test               |
+--------------------+
9 rows in set (0.052 sec)
```

2) Create two table name Employee and Department will following columns using appropriate data types and insert the following data in table.

Your table must be in following form:

Employee

emp_id	emp_name	Position	salary	dept_id
1	Anish	Manger	25000	1
2	Sita	Secretary	25000	2
3	Ronit	Analyst	40000	5
4	Riya	Manager	50000	3

Department

dept_id	dept_name	HOD
1	Sales	Janak
2	Marketing	Madhav
3	Finance	Sapana
4	Operations	Durga

```
MariaDB [lab5_dbms]> CREATE TABLE Employee(emp_id INT PRIMARY KEY,emp_name VARCHAR(50),position VARCHAR(50),salary INT,dept_id INT);
Query OK, 0 rows affected (0.014 sec)

MariaDB [lab5_dbms]> describe Employee;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| emp_id   | int(11)     | NO   | PRI | NULL    |       |
| emp_name | varchar(50) | YES  |     | NULL    |       |
| position | varchar(50) | YES  |     | NULL    |       |
| salary   | int(11)     | YES  |     | NULL    |       |
| dept_id  | int(11)     | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
5 rows in set (0.025 sec)
```

```
MariaDB [lab5_dbms]> select * from Employee;
+--------+----------+-----------+--------+---------+
| emp_id | emp_name | position  | salary | dept_id |
+--------+----------+-----------+--------+---------+
|      1 | Anish    | Manager   |  25000 |       1 |
|      2 | Sita     | Secretary |  25000 |       2 |
|      3 | Ronit    | Analyst   |  40000 |       5 |
|      4 | Riya     | Manager   |  50000 |       3 |
+--------+----------+-----------+--------+---------+
4 rows in set (0.001 sec)
```

```
MariaDB [lab5_dbms]> INSERT INTO Department VALUES (1, 'Sales', 'Janak'),(2, 'Marketing', 'Madhav'),(3, 'Finance', 'Sapana'),(4, 'Operations', 'Durga');
Query OK, 4 rows affected (0.007 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [lab5_dbms]> select * from Department;
+---------+------------+--------+
| dept_id | dept_name  | HOD    |
+---------+------------+--------+
|       1 | Sales      | Janak  |
|       2 | Marketing  | Madhav |
|       3 | Finance    | Sapana |
|       4 | Operations | Durga  |
+---------+------------+--------+
4 rows in set (0.001 sec)
```

3) Now perform the following Join Operations

1. Inner Join

```
MariaDB [lab5_dbms]> SELECT*FROM Employee INNER JOIN Department ON Employee.dept_id = Department.dept_id;
+--------+----------+-----------+--------+---------+---------+------------+--------+
| emp_id | emp_name | position  | salary | dept_id | dept_id | dept_name  | HOD    |
+--------+----------+-----------+--------+---------+---------+------------+--------+
|      1 | Anish    | Manager   | 25000  |       1 |       1 | Sales      | Janak  |
|      2 | Sita     | Secretary | 25000  |       2 |       2 | Marketing  | Madhav |
|      4 | Riya     | Manager   | 50000  |       3 |       3 | Finance    | Sapana |
+--------+----------+-----------+--------+---------+---------+------------+--------+
3 rows in set (0.001 sec)
```

2. Outer Join

➢ Left Outer Join

```
MariaDB [lab5_dbms]> SELECT * FROM Employee LEFT JOIN Department ON Employee.dept_id = Department.dept_id;
+--------+----------+-----------+--------+---------+---------+------------+--------+
| emp_id | emp_name | position  | salary | dept_id | dept_id | dept_name  | HOD    |
+--------+----------+-----------+--------+---------+---------+------------+--------+
|      1 | Anish    | Manager   | 25000  |       1 |       1 | Sales      | Janak  |
|      2 | Sita     | Secretary | 25000  |       2 |       2 | Marketing  | Madhav |
|      3 | Ronit    | Analyst   | 40000  |       5 |    NULL | NULL       | NULL   |
|      4 | Riya     | Manager   | 50000  |       3 |       3 | Finance    | Sapana |
+--------+----------+-----------+--------+---------+---------+------------+--------+
4 rows in set (0.001 sec)
```

➢ Right Outer Join

```
MariaDB [lab5_dbms]> SELECT * FROM Employee RIGHT JOIN Department ON Employee.dept_id = Department.dept_id;
+--------+----------+-----------+--------+---------+---------+------------+--------+
| emp_id | emp_name | position  | salary | dept_id | dept_id | dept_name  | HOD    |
+--------+----------+-----------+--------+---------+---------+------------+--------+
|      1 | Anish    | Manager   | 25000  |       1 |       1 | Sales      | Janak  |
|      2 | Sita     | Secretary | 25000  |       2 |       2 | Marketing  | Madhav |
|      4 | Riya     | Manager   | 50000  |       3 |       3 | Finance    | Sapana |
|   NULL | NULL     | NULL      |   NULL |    NULL |       4 | Operations | Durga  |
+--------+----------+-----------+--------+---------+---------+------------+--------+
4 rows in set (0.001 sec)
```

➢ Full Join

```
MariaDB [lab5_dbms]> SELECT * FROM Employee LEFT JOIN Department ON Employee.dept_id =Department.dept_id UNION SELECT * FROM Employee RIGHT JOIN Departmer
Employee.dept_id = Department.dept_id;
+--------+----------+-----------+--------+---------+---------+------------+--------+
| emp_id | emp_name | position  | salary | dept_id | dept_id | dept_name  | HOD    |
+--------+----------+-----------+--------+---------+---------+------------+--------+
|      1 | Anish    | Manager   | 25000  |       1 |       1 | Sales      | Janak  |
|      2 | Sita     | Secretary | 25000  |       2 |       2 | Marketing  | Madhav |
|      3 | Ronit    | Analyst   | 40000  |       5 |    NULL | NULL       | NULL   |
|      4 | Riya     | Manager   | 50000  |       3 |       3 | Finance    | Sapana |
|   NULL | NULL     | NULL      |   NULL |    NULL |       4 | Operations | Durga  |
+--------+----------+-----------+--------+---------+---------+------------+--------+
5 rows in set (0.002 sec)
```

3. Natural Join

```
MariaDB [lab5_dbms]> SELECT * FROM Employee NATURAL JOIN Department;
+---------+--------+----------+-----------+--------+------------+--------+
| dept_id | emp_id | emp_name | position  | salary | dept_name  | HOD    |
+---------+--------+----------+-----------+--------+------------+--------+
|       1 |      1 | Anish    | Manager   | 25000  | Sales      | Janak  |
|       2 |      2 | Sita     | Secretary | 25000  | Marketing  | Madhav |
|       3 |      4 | Riya     | Manager   | 50000  | Finance    | Sapana |
+---------+--------+----------+-----------+--------+------------+--------+
3 rows in set (0.001 sec)
```

4. Cross Join

```
MariaDB [lab5_dbms]> SELECT * FROM Employee CROSS JOIN Department;
+--------+----------+-----------+--------+---------+--------+-----------+--------+
| emp_id | emp_name | position  | salary | dept_id | dept_id | dept_name | HOD    |
+--------+----------+-----------+--------+---------+--------+-----------+--------+
|      1 | Anish    | Manager   |  25000 |       1 |      1 | Sales     | Janak  |
|      2 | Sita     | Secretary |  25000 |       2 |      1 | Sales     | Janak  |
|      3 | Ronit    | Analyst   |  40000 |       5 |      1 | Sales     | Janak  |
|      4 | Riya     | Manager   |  50000 |       3 |      1 | Sales     | Janak  |
|      1 | Anish    | Manager   |  25000 |       1 |      2 | Marketing | Madhav |
|      2 | Sita     | Secretary |  25000 |       2 |      2 | Marketing | Madhav |
|      3 | Ronit    | Analyst   |  40000 |       5 |      2 | Marketing | Madhav |
|      4 | Riya     | Manager   |  50000 |       3 |      2 | Marketing | Madhav |
|      1 | Anish    | Manager   |  25000 |       1 |      3 | Finance   | Sapana |
|      2 | Sita     | Secretary |  25000 |       2 |      3 | Finance   | Sapana |
|      3 | Ronit    | Analyst   |  40000 |       5 |      3 | Finance   | Sapana |
|      4 | Riya     | Manager   |  50000 |       3 |      3 | Finance   | Sapana |
|      1 | Anish    | Manager   |  25000 |       1 |      4 | Operations | Durga |
|      2 | Sita     | Secretary |  25000 |       2 |      4 | Operations | Durga |
|      3 | Ronit    | Analyst   |  40000 |       5 |      4 | Operations | Durga |
|      4 | Riya     | Manager   |  50000 |       3 |      4 | Operations | Durga |
+--------+----------+-----------+--------+---------+--------+-----------+--------+
16 rows in set (0.000 sec)
```

4) Write SQL queries to

➢ Find Employee names with their department

```
MariaDB [lab5_dbms]> SELECT emp_name, dept_name FROM Employee JOIN Department ON Employee.dept_id = Department.dept_id;
+----------+-----------+
| emp_name | dept_name |
+----------+-----------+
| Anish    | Sales     |
| Sita     | Marketing |
| Riya     | Finance   |
+----------+-----------+
3 rows in set (0.001 sec)
```

➢ Find emp_name,position,salary of employee who works in finance department

```
MariaDB [lab5_dbms]> SELECT emp_name, position, salary FROM Employee JOIN Department ON Employee.dept_id = Department.dept_id WHERE dept_name = 'Finance';
+----------+----------+--------+
| emp_name | position | salary |
+----------+----------+--------+
| Riya     | Manager  |  50000 |
+----------+----------+--------+
1 row in set (0.001 sec)
```

➢ Find emp_name of employee who works in sales department

```
MariaDB [lab5_dbms]> SELECT emp_name FROM Employee JOIN Department ON Employee.dept_id = Department.dept_id WHERE dept_name = 'Sales';
+----------+
| emp_name |
+----------+
| Anish    |
+----------+
1 row in set (0.001 sec)
```

➢ Find the information of employee who works in marketing department

```
MariaDB [lab5_dbms]> SELECT * FROM Employee JOIN Department ON Employee.dept_id = Department.dept_id WHERE dept_name = 'Marketing';
+--------+----------+-----------+--------+---------+--------+-----------+--------+
| emp_id | emp_name | position  | salary | dept_id | dept_id | dept_name | HOD    |
+--------+----------+-----------+--------+---------+--------+-----------+--------+
|      2 | Sita     | Secretary |  25000 |       2 |      2 | Marketing | Madhav |
+--------+----------+-----------+--------+---------+--------+-----------+--------+
1 row in set (0.001 sec)
```

➢ Find the HOD of Riya

```
MariaDB [lab5_dbms]> SELECT HOD FROM Employee JOIN Department ON Employee.dept_id = Department.dept_id WHERE emp_name = 'Riya';
+--------+
| HOD    |
+--------+
| Sapana |
+--------+
1 row in set (0.001 sec)
```

Problem 2:

1) Create two table named **emp_civildepartment** and **emp_computerdepartment** with following columns with appropriate data types and insert the following data in table
After insertion your table must look like as follows:

emp_civildepartment

emp_id	emp_name	address
1	Kapil	Chitwan
2	Ujwal	Dharan
3	Pradip	Palpa
4	Prakash	Kathmandu
5	Supriya	Mahendranagar

emp_computerdepartment

emp_id	emp_name	address
1	Mukunda	Surkhet
2	Santosh	Pokhara
3	Pradip	Palpa
4	Jasbin	Kathmandu
5	Supriya	Mahendranagar

```
MariaDB [lab5_dbms]> CREATE TABLE emp_civildepartment (emp_id INT, emp_name VARCHAR(50), address VARCHAR(50));
Query OK, 0 rows affected (0.017 sec)

MariaDB [lab5_dbms]> describe emp_civildepartment;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| emp_id   | int(11)     | YES  |     | NULL    |       |
| emp_name | varchar(50) | YES  |     | NULL    |       |
| address  | varchar(50) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.024 sec)
```

```
MariaDB [lab5_dbms]> INSERT INTO emp_civildepartment VALUES (1, 'Kapil', 'Chitwan'), (2, 'Ujwal', 'Dharan'), (3, 'Pradip', 'Palpa'), (4, 'Prakash', 'Kathmandu'), (5, 'Supriya', 'Mahendranagar');
Query OK, 5 rows affected (0.007 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [lab5_dbms]> select * from emp_civildepartment;
+--------+----------+---------------+
| emp_id | emp_name | address       |
+--------+----------+---------------+
|      1 | Kapil    | Chitwan       |
|      2 | Ujwal    | Dharan        |
|      3 | Pradip   | Palpa         |
|      4 | Prakash  | Kathmandu     |
|      5 | Supriya  | Mahendranagar |
+--------+----------+---------------+
5 rows in set (0.001 sec)
```

```
MariaDB [lab5_dbms]> CREATE TABLE emp_computerdepartment (emp_id INT,emp_name VARCHAR(50),address VARCHAR(50));
Query OK, 0 rows affected (0.016 sec)

MariaDB [lab5_dbms]> describe emp_computerdepartment;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| emp_id   | int(11)     | YES  |     | NULL    |       |
| emp_name | varchar(50) | YES  |     | NULL    |       |
| address  | varchar(50) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.022 sec)

MariaDB [lab5_dbms]> INSERT INTO emp_computerdepartment VALUES (1, 'Mukunda', 'Surkhet'), (2, 'Santosh', 'Pokhara'), (3, 'Pradip', 'Palpa'), (4, 'Jasbin', 'Kathmandu'), (5, 'Supriya',
'Mahendranagar');
Query OK, 5 rows affected (0.007 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [lab5_dbms]> select * from emp_computerdepartment;
+--------+----------+---------------+
| emp_id | emp_name | address       |
+--------+----------+---------------+
|      1 | Mukunda  | Surkhet       |
|      2 | Santosh  | Pokhara       |
|      3 | Pradip   | Palpa         |
|      4 | Jasbin   | Kathmandu     |
|      5 | Supriya  | Mahendranagar |
+--------+----------+---------------+
5 rows in set (0.001 sec)
```

2) perform the following set operations

➢ UNION

```
MariaDB [lab5_dbms]> SELECT* FROM emp_civildepartment UNION SELECT *FROM emp_computerdepartment;
+--------+----------+---------------+
| emp_id | emp_name | address       |
+--------+----------+---------------+
|      1 | Kapil    | Chitwan       |
|      2 | Ujwal    | Dharan        |
|      3 | Pradip   | Palpa         |
|      4 | Prakash  | Kathmandu     |
|      5 | Supriya  | Mahendranagar |
|      1 | Mukunda  | Surkhet       |
|      2 | Santosh  | Pokhara       |
|      4 | Jasbin   | Kathmandu     |
+--------+----------+---------------+
8 rows in set (0.000 sec)
```

➢ UNION ALL

```
MariaDB [lab5_dbms]> SELECT* FROM emp_civildepartment UNION ALL SELECT *FROM emp_computerdepartment;
+--------+----------+---------------+
| emp_id | emp_name | address       |
+--------+----------+---------------+
|      1 | Kapil    | Chitwan       |
|      2 | Ujwal    | Dharan        |
|      3 | Pradip   | Palpa         |
|      4 | Prakash  | Kathmandu     |
|      5 | Supriya  | Mahendranagar |
|      1 | Mukunda  | Surkhet       |
|      2 | Santosh  | Pokhara       |
|      3 | Pradip   | Palpa         |
|      4 | Jasbin   | Kathmandu     |
|      5 | Supriya  | Mahendranagar |
+--------+----------+---------------+
10 rows in set (0.000 sec)
```

➢ INTERSET

```
MariaDB [lab5_dbms]> SELECT* FROM emp_civildepartment UNION ALL SELECT *FROM emp_computerdepartment;
+--------+----------+---------------+
| emp_id | emp_name | address       |
+--------+----------+---------------+
|      1 | Kapil    | Chitwan       |
|      2 | Ujwal    | Dharan        |
|      3 | Pradip   | Palpa         |
|      4 | Prakash  | Kathmandu     |
|      5 | Supriya  | Mahendranagar |
|      1 | Mukunda  | Surkhet       |
|      2 | Santosh  | Pokhara       |
|      3 | Pradip   | Palpa         |
|      4 | Jasbin   | Kathmandu     |
|      5 | Supriya  | Mahendranagar |
+--------+----------+---------------+
10 rows in set (0.000 sec)
```

➢ EXCEPT

```
MariaDB [lab5_dbms]> SELECT * FROM emp_computerdepartment WHERE emp_name NOT IN (SELECT emp_name FROM emp_civildepartment);
+--------+----------+------------+
| emp_id | emp_name | address    |
+--------+----------+------------+
|      1 | Mukunda  | Surkhet    |
|      2 | Santosh  | Pokhara    |
|      4 | Jasbin   | Kathmandu  |
+--------+----------+------------+
3 rows in set (0.001 sec)
```

3) Write SQL queries for the following:

➢ Find the name of employee who works either civil or computer department

```
MariaDB [lab5_dbms]> SELECT emp_name FROM emp_civildepartment UNION SELECT emp_name FROM emp_computerdepartment;
+----------+
| emp_name |
+----------+
| Kapil    |
| Ujwal    |
| Pradip   |
| Prakash  |
| Supriya  |
| Mukunda  |
| Santosh  |
| Jasbin   |
+----------+
8 rows in set (0.001 sec)
```

➢ Find the name of employee who works in both civil and computer department

```
MariaDB [lab5_dbms]> SELECT c.emp_name FROM emp_civildepartment c JOIN emp_computerdepartment d ON c.emp_name = d.emp_name;
+----------+
| emp_name |
+----------+
| Pradip   |
| Supriya  |
+----------+
2 rows in set (0.001 sec)
```

➢ Find the name of employee who works in computer department but not in civil department.

```
MariaDB [lab5_dbms]> SELECT emp_name FROM emp_computerdepartment WHERE emp_name NOT IN (SELECT emp_name FROM emp_civildepartment);
+----------+
| emp_name |
+----------+
| Mukunda  |
| Santosh  |
| Jasbin   |
+----------+
3 rows in set (0.001 sec)
```

➢ Find the common address of where both computer and civil department employee lives in .

```
MariaDB [lab5_dbms]> SELECT DISTINCT c.address FROM emp_civildepartment c JOIN emp_computerdepartment d ON c.address = d.address;
+----------------+
| address        |
+----------------+
| Palpa          |
| Kathmandu      |
| Mahendranagar  |
+----------------+
3 rows in set (0.001 sec)

MariaDB [lab5_dbms]>
```